

ChE 381: Unit Operations Laboratory

FLUID FLOW IN PIPE SYSTEMS

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Introduction

Most chemical processes involve pipe systems to transport materials to and from various reactors, separation process, and storage facilities. This can involve single-phase fluids (gases or liquids) and solids or two-phase flows (e.g., slurry of coal in water.)

An important part of chemical engineering design is the analysis of these flow systems. It is necessary to calculate the pressure drop in various sections to be able to rationally select a pump to produce the desired flow. At times, this can be a difficult task involving both theoretical analysis and practical well-established empirical correlations.

The purpose of this experiment is to design a set of experiments to obtain energy loss coefficients such as friction factor, f , loss coefficient, K , and flow coefficient, C_f , on various parts of the fluid-flow system in the Unit Operations Laboratory, and to use these data to test the available theories and correlations for fluid flow in pipes and fittings.

Experimental Equipment

The fluid flow system consists of a Hampden **H-6920** Pipe Friction Demonstrator; a bench mounted panel upon which is located the flow circuits and instrumentation, as well as a separate pump, reservoir tank and rotameter to control the flow of water into and out of the system. Further details are given below and also in the H-6920 Operating Instructions/Experimental Manual (located in the lab).

The flow circuit is comprised of one main path with nine parallel branches at the top. The quick-connect water plugs are located in the lower right corner. The shut-off valves are located in the upper left corner and the flow control valves are located in the lower right corner.

The pipe lengths used for the pipe friction head loss experiments are the nine parallel branches at the top consisting of copper tubing, plastic pipe and stainless steel pipe. The characteristics of the piping are given in the Table below, where k_s is the relative roughness for the pipe:

PIPE MATERIAL	k_s (NOMINAL VALUE)	SIZE	INNER DIAMETER
Copper tubing	$5 * 10^{-6}$ ft ($1.5 * 10^{-3}$ mm)	1/2"	0.402" (10.21 mm)
		3/4"	0.666" (16.92 mm)
		1-1/8"	1.025" (26.04 mm)
Plastic Pipe	smooth	3/8" NPT	0.493" (12.52 mm)
		1/2" NPT	0.622" (15.80 mm)
		1" NPT	1.049" (26.64 mm)
Stainless Steel Pipe	$1.5 * 10^{-4}$ ft ($4.6 * 10^{-2}$ mm)	3/8" NPT	0.493" (12.52 mm)
		1/2" NPT	0.622" (15.80 mm)
		1" NPT	1.049" (26.64 mm)

The fittings used for the pipe fitting head loss experiments and the orifice flow meter for the flow measurement experiments are found in the leg across the middle and lower section of the panel. The orifice flowmeter has an installed orifice plate of 0.478" (12.1mm). The fittings consist of pipe sections containing a 180° Bend, a 90° Short radius Elbow, and a section of pipe with an expansion followed by a contraction. Pressure taps (sensing ports) allow determination of the appropriate pressure drops across the straight pipe sections, fittings and orifice flowmeter.

The instrumentation is found in the center of the panel. It consists of a thermocouple and digital readout meter for the temperature, plus two digital differential pressure transmitters (DPT) and two manual pressure gauges to determine pressure drops over various sections of the flow system.

Experimental Studies

The goal is to obtain sufficient experimental data to allow a comparison between theory and experiment for each section of the fluid-flow system. The following experiments should be performed:

1. Rotameter Calibration. The markings on the rotameter indicate that it is calibrated for standard cubic feet per minute of air (SCFM), not liters per minute (LPM) of water. A conversion factor on the rotameter converts the reading to LPM water. Perform a set of experiments to determine that this conversion factor is correct and has not changed.
2. Orifice Meter: Experimentally determine C_f , the flow coefficient (or orifice coefficient) for the orifice flowmeter. Measured variables are the differential pressure across the orifice plate, volumetric flow rate (LPM) and water temperature. Is C_f constant, or does it vary with flowrate?
3. Pipe Friction Head Loss: Experimentally determine friction factors as a function of Reynolds number and compare resulting data with published or empirical data. Measured variables are differential pressure drop over the straight length of piping, volumetric flow rate and water temperature. Questions: Can this study cover both laminar and turbulent flow? What is the difference between the Fanning friction factor and the Darcy friction factor?
4. Pipe Fitting Head Loss: Determine the loss coefficient, K , for the pipe fittings (bends, elbows, expansion, and contraction) as a function of Reynolds number. The differential flowmeter will not be used. Measured variables are differential pressure, ΔP , volumetric flowrate, and water temperature. Question: What is the equivalent length of diameter for the fittings?

Pressure Drop Measurements:

- The pressure drop across a length of pipe, fitting or the orifice plate is measured by connecting the pressure taps, or ports on the piping or fittings to either the differential pressure transmitter (DPT) or the two manual pressure gauges via plastic tubes with quick-disconnect fittings on each end. The upstream pressure sensing port is connected to the high side of the DPT and the downstream pressure sensing port to the low side of the DPT. The differential pressure transmitters are connected to digital meters that give a direct reading of pressure drop in Δpsi . The pressure transmitters are calibrated and should give a reading close to zero (0.00 to 0.02) before connecting the pressure taps. If

the DPT zero reading is not correct, see the TA or instructor for re-zeroing. If the manual pressure gauges are used, ΔP is the difference between the two gauge readings.

Recommended procedure when switching from one measurement to another:

- Close valve to stop water flow.
- Pull out the quick disconnect fittings from the sensing ports.
- Connect into sensing ports on the next section to be measured, and open valve to start water flow.
- Flush out DPT by opening the two black valves on the side of the DPT.
- Close the black valves on the DPT
- Wait for flow in pipe/fitting to become steady, and then read the pressure drop directly from the DPT meter.

Note: The H-6920 Pipe Friction Demonstrator-Experiment Manual is a good reference for additional information on conducting the experiments.

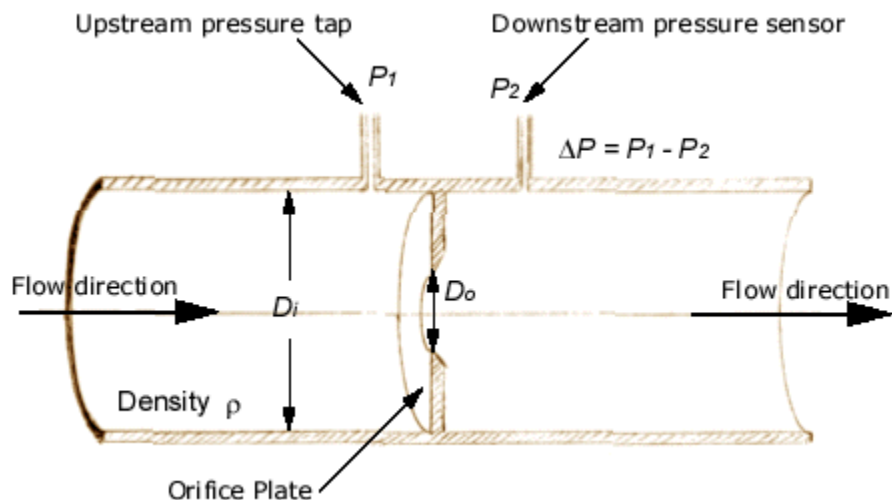
Data Analysis

Fluid flow in pipe systems has been extensively studied by chemical engineers due to its central importance to the rational design of lab-scale, pilot plant, and commercial units. There are numerous theories and correlations in the literature. The Reference Section gives several of the key sources of information which are useful in deciding how to analyze the data from the fluid-flow system experiments.

Reference Section

ORIFICE METERS: FLOW MEASUREMENT [1, 2]

A fluid passing through an orifice constriction will experience a drop in pressure across the orifice. This change can be used to measure the flowrate of the fluid.



As long as the fluid speed is sufficiently subsonic ($V < \text{mach } 0.3$), the incompressible Bernoulli's equation describes the flow reasonably well. Applying this equation to a streamline traveling down the axis of the horizontal tube gives,

$$\Delta p = p_1 - p_2 = \frac{1}{2} \rho V_2^2 - \frac{1}{2} \rho V_1^2$$

where location 1 is upstream of the orifice, and location 2 is slightly behind the orifice. It is recommended that location 1 be positioned one pipe diameter upstream of the orifice, and location 2 be positioned one-half pipe diameter downstream of the orifice. Since the pressure at 1 will be higher than the pressure at 2 (for flow moving from 1 to 2), the pressure difference as defined will be a positive quantity.

From continuity, the velocities can be replaced by cross-sectional areas of the flow and the volumetric flowrate Q ,

$$\Delta p = \frac{1}{2} \rho Q^2 \frac{1}{A_2^2} \left[1 - \left(\frac{A_2}{A_1} \right)^2 \right]$$

Solving for the volumetric flowrate Q gives,

$$Q = \sqrt{\frac{2\Delta p}{\rho}} \frac{A_2}{\sqrt{1 - \left(\frac{A_2}{A_1} \right)^2}}$$

The above equation applies only to perfectly laminar, inviscid flows. For real flows (such as water or air), viscosity and turbulence are present and act to convert kinetic flow energy into heat. To account for this effect, a **discharge coefficient** C_d is introduced into the above equation to marginally reduce the flowrate Q ,

$$Q = C_d \sqrt{\frac{2\Delta p}{\rho}} \frac{A_2}{\sqrt{1 - \left(\frac{A_2}{A_1} \right)^2}} \quad (1)$$

Since the actual flow profile at location 2 downstream of the orifice is quite complex, thereby making the effective value of A_2 uncertain, the following substitution introducing a **flow coefficient** C_f is made,

$$C_f A_o = C_d \frac{A_2}{\sqrt{1 - \left(\frac{A_2}{A_1} \right)^2}} \quad 2)$$

where A_o is the area of the orifice. As a result, the volumetric flowrate Q for real flows is given

by the equation,

$$Q = C_f A_o \sqrt{\frac{2\Delta p}{\rho}} \quad (3)$$

The flow coefficient C_f is found from experiments and is also tabulated in reference books; it ranges from 0.6 to 0.9 for most orifices. Since it depends on the orifice and pipe diameters (as well as the Reynolds Number) one will often find C_f tabulated versus the ratio of orifice diameter to inlet diameter, sometimes defined as β ,

$$\beta = \frac{D_o}{D_{inlet}} \quad (4)$$

The mass flowrate can be found by multiplying Q with the fluid density,

$$Q_{mass} = \rho Q \quad (5)$$

For more information on orifice meters, see pp. 227-231 in [1].

PIPE FRICTION HEAD LOSS

For flow in straight pipes, the most widely used correlation is the friction factor, f . Friction factor is sometimes called the resistance coefficient. The friction factor can be experimentally found if the pressure drop, ΔP , is known (Eqn 5.8 in McCabe, Smith, Harriott):

$$f = \frac{\Delta P D}{2L\rho V^2} \quad (6)$$

The head loss, h_L due to pipe friction can be calculated if the friction factor, f is known:

$$h_L = \frac{\Delta P}{\rho} = f \frac{L}{D} \frac{V^2}{2} \quad (7)$$

In laminar flow, the resistance is due to viscous forces only and is independent of the roughness of the pipe surface. It can be shown that for laminar flow, the friction factor is given by:

$$f = \frac{64}{Re} \quad (8)$$

The friction factor for laminar flow in pipes is shown in Figure 5.10 [1]. In turbulent flow, the friction factor is due to both viscous forces and smoothness of the conduit walls. The viscous forces are given by the Reynolds number. The smoothness of the conduit is given by k_s/D where k_s is the equivalent sand roughness of the pipe. The friction factor for turbulent flow can be empirically calculated from the Colebrook equation or can be found from the Moody chart (Figure 5.10).

PIPE FITTING HEAD LOSS

A piping system includes many appurtenances such as bends, contractions, expansions, tees, couplings, etc. that result in a net loss of energy in the flowing fluid. Fluid pressure is the source of the energy loss. This energy loss is mainly caused by flow separation and additional turbulence that is present, usually observed as swirling.

The head loss due to pipe fittings can be estimated from empirical data on the subject. Due to the nature of commercially available pipe fittings, the empirical data should be used as an estimate only. In general, the head loss is given by (see Eqns 5.70 and 5.72, McCabe, Smith, Harriot):

$$h_L = K \frac{v^2}{2} \quad (9)$$

Where the loss coefficient, K, is equal to $K = f \frac{L}{D}$ (Equation 7).

The head loss in pipe fittings can also be given in terms of equivalent length of diameters, L. In this case, the pressure loss in a piping system can be estimated by adding together the equivalent lengths of all the fittings and the straight pipe length and determining a loss coefficient for the total piping system. See Reference [1] pp. 121-128 for more information.

REFERENCES

- [1] W. McCabe, J. Smith, and P. Harriot, Unit Operations of Chemical Engineering, Chapter 5 and Chapter 8 pp 224-241, 7th ed. McGraw-Hill, 2005.
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- [3] R. H. Perry and D. W. Green, Eds., Perry's Chemical Engineer's Handbook, 7th ed. McGraw-Hill, 1997.
- [4] A. S. Foust, Principles of Unit Operations, 2nd ed. John Wiley and Sons, Inc., 1980.